

Kam To Billy SIEVERS

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Theoretical Physics Ph.D. Candidate exploring opportunities in the financial sector. Expert in Monte Carlo sampling and spectral analysis in Python and C. Recipient of the NSERC PGS-D scholarship with a track record of research publications and presentations.

HIGHLIGHTS OF QUALIFICATIONS

- Ph.D. Candidate in Physics graduating August 2027
- Traveled award-winning presenter of cutting-edge black hole physics research
- Four articles published in reputable black hole physics journals, with more manuscripts on the way

EDUCATION

McMaster University Hamilton, ON
Doctor of Philosophy (Ph.D.) in Physics 2023 – present
Recipient of NSERC PGS-D Scholarship (\$101 000 total over 3 years). Expected graduation in 2027.

Memorial University of Newfoundland St. John's, NL
Master of Science (M.Sc.) in Physics 2020 – 2023
GPA: 4.00 / 4.00 | Fellow of the School of Graduate Studies.

University of Calgary Calgary, AB
Bachelor of Science (B.Sc.) Honours in Physics 2016 – 2020
First Class Honours | GPA: 3.60 / 4.00
Recipient of International Student Scholarship (\$60 000 total over 4 years).

TECHNICAL PRESENTATIONS

- **International Research Seminar Tour (2022):**
Invited to present numerical results [1] at the Max Planck Institute (Germany), Institute of Cosmos Sciences of the University of Barcelona (Spain), and Institute of Theoretical Physics at Charles University (Czechia). Each visit lasted one week and each seminar presentation was one hour long.
- **Conference Presentations (2021–present):**
Delivered over 10 talks (~15 minutes each) at national conferences, including Theory Canada and CAP Congress. Awarded 1st place for Best M.Sc. Talk and 2nd place for Best Student Talk at the Atlantic General Relativity Meeting (2021, 2022).

TECHNICAL SKILLS

Numerical Methods: Monte Carlo Simulations, Finite Differencing, (Pseudo-)Spectral Methods.

Programming: Python, C, Git, Linux.

Others: Teaching (9+ years), international seminar and conference presentations (6+ years).

PUBLICATIONS

[1] *Physical Review D*, **105**, 044024 (2022): Interior marginally outer trapped surfaces of spherically symmetric black holes.
Formulated a physical problem into an quantified optimization problem, then solved it by developing a custom Monte Carlo program leading to exotic findings within the context of a black hole.

[2] *Classical and Quantum Gravity*, **40**, 095010 (2023): Exotic surfaces in rotating spacetimes.
Further generalized previously developed Monte Carlo pipeline on an even more difficult physical system – the model of a rotating black hole.

[3] *Physical Review D*, **109**, 124023 (2024): Marginally Outer Trapped Tori in Black Hole Spacetimes.

[4] *Journal of Physics A*, **58**, 405204 (2025): Spectrum of the Laplacian on the Page metric.
Led the implementation of a pseudo-spectral method to numerically approximate eigenvalues to an open geometry problem.

PROFESSIONAL EXPERIENCE AND AFFILIATIONS

Employment	
2020–present	McMaster University & Memorial University <i>Graduate Research and Teaching Assistant</i> In addition to research and teaching duties, also mentors students in my research groups, leading to three publications [1, 3, 4] and two undergraduate theses.
2024–present	Billy’s Squash Shop , Hamilton ON <i>Founder and Sole Proprietor</i>
2017–2023	MathPro Learning Centre , Calgary AB <i>Team-Lead & Instructor</i> Developed an extracurricular coding course in Python, tutored high-school students one-on-one in calculus and physics (Alberta curriculum).
Volunteering	
2025–present	McMaster Astronomy and Physics Graduate Students Association <i>President</i>
2025–present	McMaster Mens Squash Team <i>Equipment Manager & Player</i>
Affiliations	
2025–present	Perimeter Institute for Theoretical Physics , Waterloo ON <i>Associate Graduate Student</i>
2025–present	Digital Research Alliance of Canada <i>Active member – Doctoral Student</i>
2023–present	Classical and Quantum Gravity <i>Referee</i>
2021–present	Canadian Association of Physicists <i>Graduate Student Member</i>